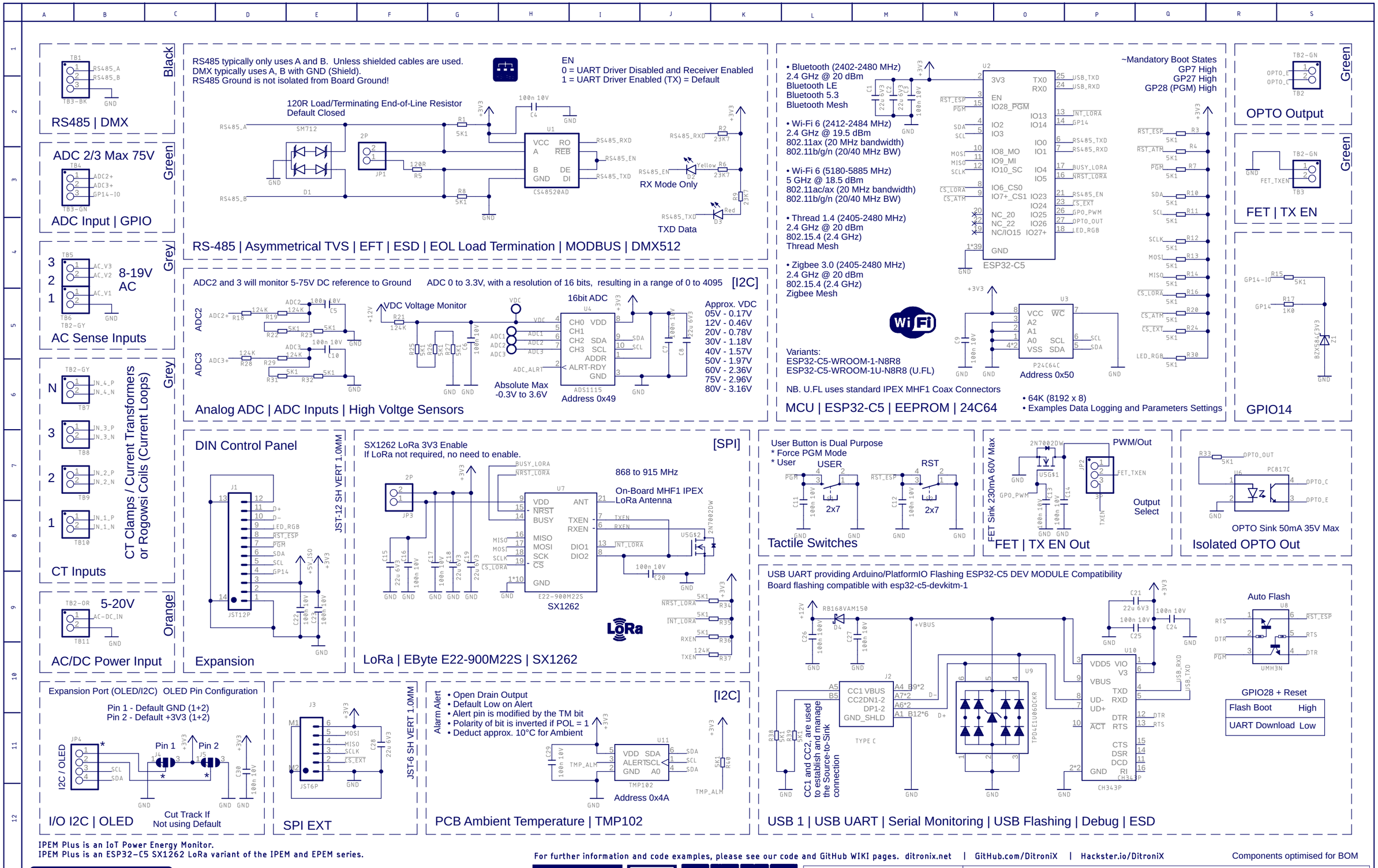
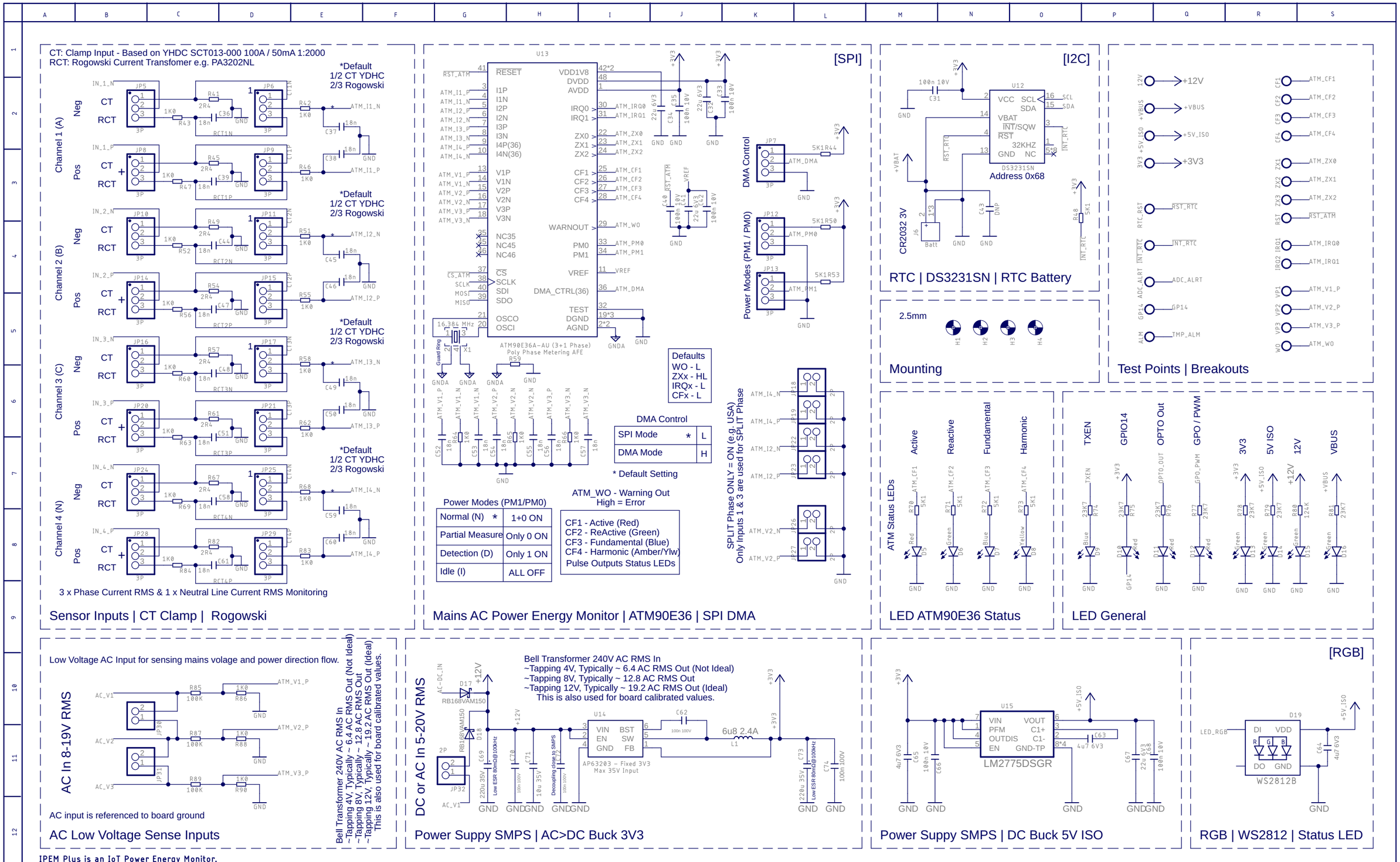

RS485 | DMX
Black
RS485 typically only uses A and B. Unless shielded cables are used.
DMX typically uses A, B with GND (Shield).
RS485 Ground is not isolated from Board Ground!
EN
0 = UART Driver Disabled and Receiver Enabled
1 = UART Driver Enabled (TX) = Default
120R Load/Terminating End-of-Line Resistor
Default Closed
RX Mode Only
TXD Data
RS-485 | Asymmetrical TVS | EFT | ESD | EOL Load Termination | MODBUS | DMX512
ADC 2/3 Max 75V
Green
ADC Input | GPIO
• Bluetooth (2402-2480 MHz)
2.4 GHz @ 20 dBm
Bluetooth LE
Bluetooth 5.3
Bluetooth Mesh
• Wi-Fi 6 (2412-2484 MHz)
2.4 GHz @ 19.5 dBm
802.11ax (20 MHz bandwidth)
802.11b/g/n (20/40 MHz BW)
• Wi-Fi 6 (5180-5885 MHz)
5 GHz @ 18.5 dBm
802.11ac/ax (20 MHz bandwidth)
802.11b/g/n (20/40 MHz BW)
• Thread 1.4 (2405-2480 MHz)
2.4 GHz @ 20 dBm
802.15.4 (2.4 GHz)
Thread Mesh
• Zigbee 3.0 (2405-2480 MHz)
2.4 GHz @ 20 dBm
802.15.4 (2.4 GHz)
Zigbee Mesh
~Mandatory Boot States
GP7 High
GP27 High
GP28 (PGM) High
ESP32-C5
Variants:
ESP32-C5-WROOM-1-N8R8
ESP32-C5-WROOM-1U-N8R8 (U.FL)
NB. U.FL uses standard IPEX MHF1 Coax Connectors
Address 0x50
• 64K (8192 x 8)
• Examples Data Logging and Parameters Settings
MCU | ESP32-C5 | EEPROM | 24C64
OPTO Output
Green
FET | TX EN
Green
GPIO14
8-19V
AC
Grey
AC Sense Inputs
ADC2 and 3 will monitor 5-75V DC reference to Ground
ADC 0 to 3.3V, with a resolution of 16 bits, resulting in a range of 0 to 4095
[I2C]
VDC Voltage Monitor
16bit ADC
Approx. VDC
05V - 0.17V
12V - 0.46V
20V - 0.78V
30V - 1.18V
40V - 1.57V
50V - 1.97V
60V - 2.36V
75V - 2.96V
80V - 3.16V
Absolute Max
-0.3V to 3.6V
Address 0x49
Analog ADC | ADC Inputs | High Voltge Sensors
CT Clamps / Current Transformers
or Rogowsi Coils (Current Loops)
Grey
CT Inputs
5-20V
Orange
AC/DC Power Input
DIN Control Panel
JST-12 SH VERT 1.0MM
Expansion
SX1262 LoRa 3V3 Enable
If LoRa not required, no need to enable.
[SPI]
868 to 915 MHz
On-Board MHF1 IPEX
LoRa Antenna
SX1262
LoRa | EByte E22-900M22S | SX1262
User Button is Dual Purpose
* Force PGM Mode
* User
USER
RST
2x7
Tactile Switches
FET Sink 230mA 60V Max
PWM/Out
Output
Select
FET | TX EN Out
OPTO Sink 50mA 35V Max
Isolated OPTO Out
USB UART providing Arduino/PlatformIO Flashing ESP32-C5 DEV MODULE Compatibility
Board flashing compatible with esp32-c5-devkitm-1
Auto Flash
GPIO28 + Reset
Flash Boot High
UART Download Low
CC1 and CC2, are used
to establish and manage
the Source-to-Sink
connection
USB 1 | USB UART | Serial Monitoring | USB Flashing | Debug | ESD
Expansion Port (OLED/I2C) OLED Pin Configuration
Pin 1 - Default GND (1+2)
Pin 2 - Default +3V3 (1+2)
Pin 1
Pin 2
Cut Track If
Not using Default
I/O I2C | OLED
JST-6 SH VERT 1.0MM
SPI EXT
[I2C]
Alarm Alert
• Open Drain Output
• Default Low on Alert
• Alert pin is modified by the TM bit
• Polarity of bit is inverted if POL = 1
• Deduct approx. 10°C for Ambient
Address 0x4A
PCB Ambient Temperature | TMP102
IPEM Plus is an IoT Power Energy Monitor.
IPEM Plus is an ESP32-C5 SX1262 LoRa variant of the IPEM and EPEM series.
For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX
Components optimised for BOM
DitroniX
Home Automation | Internet of Things | Energy Monitoring
Production
(C)2018-2026 Dave Williams | DitroniX | G8PUO
Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8
STEM
ESP32-C5 SX1262 SDK
ditronix.net
DitroniX Ltd – UK
IPEM Plus ESP32-C5 SX1262 E22-900M22S LoRa 1.2605.103
ATM90E36-ESP32-C5-LORA-IOT-POWER-ENERGY-MONITOR-BOARD-(PG1)
Sheet: 1/2
Drwg/Chk: Dave Williams
06/05/2026 07:32



IPEM Plus is an IoT Power Energy Monitor.
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Components optimised for BOM

DitroniX Home Automation | Internet of Things | Energy Monitoring

Production

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Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8

ESP32-C5 SX1262 SDK ditronix.net DitroniX Ltd - UK	IPEM Plus ESP32-C5 SX1262 E22-900M22S LoRa 1.2605.103
	ATM90E36-ESP32-C5-LORA-IOT-POWER-ENERGY-MONITOR-BOARD-(PG2)
	Sheet: 2/2 Drwg/Chk: Dave Williams 06/05/2026 07:32